ASSIGNMENT 1

Problem Statement :

Write a c program to check whether the year is a leap year or not.

Code :

```c
#include <stdio.h>
int main() {
   int year;
   printf("Enter a year: ");
   scanf("%d", &year);

   // leap year if perfectly divisible by 400
   if (year % 400 == 0) {
      printf("%d is a leap year.", year);
   }
   // not a leap year if divisible by 100
   // but not divisible by 400
   else if (year % 100 == 0) {
      printf("%d is not a leap year.", year);
   }
   // leap year if not divisible by 100
   // but divisible by 4
   else if (year % 4 == 0) {
      printf("%d is a leap year.", year);
   }
   // all other years are not leap years
   else {
      printf("%d is not a leap year.", year);
   }

   return 0;
}
```

Output :

```
Enter a year: 1900
1900 is not a leap year.
```

ASSIGNMENT 2

Problem Statement :

Write a menu driven program in C to implement the base arithmetic operations such as Addition, Substraction, Multiplication and Division

Code :

```c
#include <stdio.h>  // Include the standard input/output header file.

void main() {
  int num1,num2,opt;  // Declare variables to store user input and operation choice
      .

  // Prompt user for two integers and store them.
  printf("Enter the first Integer :");
  scanf("%d",&num1);
  printf("Enter the second Integer :");
  scanf("%d",&num2);

  // Display the menu for operation choice.
  printf("\nInput your option :\n");
  printf("1-Addition.\n2-Substraction.\n3-Multiplication.\n4-Division.\n5-Exit.\n"
      );
  scanf("%d",&opt);  // Read and store the user's choice.

  switch(opt) {  // Start a switch statement based on the user's choice.
    case 1:
      printf("The Addition of  %d and %d is: %d\n",num1,num2,num1+num2);  //
          Perform addition and print result.
      break;

    case 2:
      printf("The Substraction of %d  and %d is: %d\n",num1,num2,num1-num2);  //
          Perform subtraction and print result.
      break;
```

```c
    case 3:
      printf("The Multiplication of %d  and %d is: %d\n",num1,num2,num1*num2);  //
          Perform multiplication and print result.
      break;

    case 4:
      if(num2==0) {
        printf("The second integer is zero. Divide by zero.\n");  // Handle
            division by zero.
      } else {
        printf("The Division of %d  and %d is : %d\n",num1,num2,num1/num2);  //
            Perform division and print result.
      }
      break;

    case 5:
      break;  // Exit the program.

    default:
      printf("Input correct option\n");  // Display error message for invalid input
          .
      break;
  }
}
```

Output :

```
Enter the first Integer :2
Enter the second Integer :3

Input your option :
1-Addition.
2-Substraction.
3-Multiplication.
4-Division.
5-Exit.
3
The Multiplication of 2  and 3 is: 6
```

ASSIGNMENT 3

Problem Statement :

Write a program in C to generate multiplication tables for a given number.

Code :

```c
#include <stdio.h>
int main() {
  int n;
  printf("Enter an integer: ");
  scanf("%d", &n);

  for (int i = 1; i <= 10; ++i) {
    printf("%d * %d = %d \n", n, i, n * i);
  }
  return 0;
}
```

Output :

```
Enter an integer: 9
9 * 1 = 9
9 * 2 = 18
9 * 3 = 27
9 * 4 = 36
9 * 5 = 45
9 * 6 = 54
9 * 7 = 63
9 * 8 = 72
9 * 9 = 81
9 * 10 = 90
```

ASSIGNMENT 4

Problem Statement :

Write a C program to calculate the salary of an employee given his basic pay. Calculate gross salary of employee. Let HRA be 10% of basic pay and TA be 5% of basic pay. Let employees pay professional tax as 2% of total salary. Calculate net salary payable after deduction.

Code :

```c
#include<stdio.h>

int main() {
    float basic_salary, gross_salary, HRA, DA;

    printf("Enter the basic salary: ");
    scanf("%f", &basic_salary);

    if(basic_salary <= 10000) {
        HRA = basic_salary * 0.2;
        DA = basic_salary * 0.8;
    } else if(basic_salary <= 20000) {
        HRA = basic_salary * 0.25;
        DA = basic_salary * 0.9;
    } else {
        HRA = basic_salary * 0.3;
        DA = basic_salary * 0.95;
    }

    gross_salary = basic_salary + HRA + DA;
    printf("Gross Salary is: %.2f\n", gross_salary);

    return 0;
}
```

Output :

```
Enter the basic salary: 15000
Gross Salary is: 32250.00
```

ASSIGNMENT 5

Problem Statement :

Write a program in C to perform basic matrix operations such as:

1. Addition of two matrices

2. Saddle point of a matrix

3. Inverse of a matrix

4. Magic square of a matrix

Code :

```c
// Matrix Addition
#include <stdio.h>

#define ROW 3
#define COL 3

// Function to print a matrix
void printMatrix(int matrix[ROW][COL]) {
    for (int i = 0; i < ROW; i++) {
        for (int j = 0; j < COL; j++) {
            printf("%d ", matrix[i][j]);
        }
        printf("\n");
    }
}

int main() {
    // Step 1: Define matrices A and B
    int matrixA[ROW][COL] = {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}
    };

    int matrixB[ROW][COL] = {
        {9, 8, 7},
        {6, 5, 4},
        {3, 2, 1}
    };
```

Output :

```
Matrix A:
1 2 3
4 5 6
7 8 9

Matrix B:
9 8 7
6 5 4
3 2 1

Performing Matrix Addition (A + B):
Sum Matrix:
10 10 10
10 10 10
10 10 10
```

ASSIGNMENT 6

Problem Statement :

Write a C function to compute the factorial of a number with and without recursion.

Code :

```c
#include <stdio.h>
int main() {
    int n, i;
    unsigned long long fact = 1;
    printf("Enter an integer: ");
    scanf("%d", &n);

    // shows error if the user enters a negative integer
    if (n < 0)
        printf("Error! Factorial of a negative number doesn't exist.");
    else {
        for (i = 1; i <= n; ++i) {
            fact *= i;
        }
        printf("Factorial of %d = %llu", n, fact);
    }

    return 0;
}
```

Output :

```
Enter an integer: 10
Factorial of 10 = 3628800
```

ASSIGNMENT 7

Problem Statement :

Write a C program to accept student details and display their result using an array of structures

Code :

```c
#include <stdio.h>
struct student {
    char firstName[50];
    int roll;
    float marks;
} s[5];

int main() {
    int i;
    printf("Enter information of students:\n");

    // storing information
    for (i = 0; i < 5; ++i) {
        s[i].roll = i + 1;
        printf("\nFor roll number%d,\n", s[i].roll);
        printf("Enter first name: ");
        scanf("%s", s[i].firstName);
        printf("Enter marks: ");
        scanf("%f", &s[i].marks);
    }
    printf("Displaying Information:\n\n");
```

```c
    // displaying information
    for (i = 0; i < 5; ++i) {
        printf("\nRoll number: %d\n", i + 1);
        printf("First name: ");
        puts(s[i].firstName);
        printf("Marks: %.1f", s[i].marks);
        printf("\n");
    }
    return 0;
}
```

Output :

```
Enter information of students:

For roll number1,
Enter name: Tom
Enter marks: 98

For roll number2,
Enter name: Jerry
Enter marks: 89
.
.
.
Displaying Information:

Roll number: 1
Name: Tom
Marks: 98
.
.
.
```

ASSIGNMENT 8

Problem Statement :

Write a program in C to accept marks of 5 courses of a student and compute the result. A student is considered PASS if he/she scores 40 marks or above in each course. If the student passes, calculate the aggregate percentage and assign the grades as follows :

1) >= 75% - Distinction

2) >= 60% - First division

3) >= 50% - Second division

4) >= 40% - Third division

Code :

```c
// C Program to Find Grade of a Student Using If else Ladder
#include <stdio.h>

int main() {
    float subject_1, subject_2, subject_3, subject_4, subject_5;
    float total, average, percentage;
    char grade;

    printf("Enter the marks of five subjects::\n");
    scanf("%f%f%f%f%f", &subject_1, &subject_2, &subject_3, &subject_4, &subject_5
        );

    // It will calculate the Total, Average and Percentage
    total = subject_1 + subject_2 + subject_3 + subject_4 + subject_5;
    average = total / 5.0;
    percentage = (total / 500.0) * 100;

    // It will calculate the Grade
    if (average >= 90)
        grade = 'A';
    else if (average >= 80 && average < 90)
        grade = 'B';
    else if (average >= 70 && average < 80)
        grade = 'C';
    else if (average >= 60 && average < 70)
        grade = 'D';
    else
        grade = 'E';
```

```
    // It will produce the final output
    printf("\nThe Total marks is:  \t%.2f / 500.00\n", total);
    printf("\nThe Average marks is:\t%.2f\n", average);
    printf("\nThe Percentage is:   \t%.2f%%\n", percentage);
    printf("\nThe Grade is:        \t'%c'\n", grade);
    return 0;
}
```

Output :

```
Enter the marks of five subjects::
90
80
70
60
50

The Total marks is:     350.00 / 500.00

The Average marks is:   70.00

The Percentage is:      70.00%

The Grade is:           'C'
```

ASSIGNMENT 9

Problem Statement :

Write a program in C to check whether the given number is an Armstrong number or not. An Armstrong number is an integer with 3 digits such that the sum of the cubes of its digits is equal to the number itself.

Code :

```c
#include <stdio.h>
int main() {
    int num, originalNum, remainder, result = 0;
    printf("Enter a three-digit integer: ");
    scanf("%d", &num);
    originalNum = num;

    while (originalNum != 0) {
       // remainder contains the last digit
        remainder = originalNum % 10;

       result += remainder * remainder * remainder;

       // removing last digit from the orignal number
       originalNum /= 10;
    }

    if (result == num)
        printf("%d is an Armstrong number.", num);
    else
        printf("%d is not an Armstrong number.", num);

    return 0;
}
```

Output :

```
Enter a three-digit integer: 371
371 is an Armstrong number.
```

ASSIGNMENT 10

Problem Statement :

Write a program in C to simulate a simple calculator that performs basic arithmetic operations such as Addition, Substraction, Multiplication and Division. The calculator should also perform special operations like computing x^y (power) and x! (factorial).

Code :

```c
#include <stdio.h>

int main() {

  char op;
  double first, second;
  printf("Enter an operator (+, -, *, /): ");
  scanf("%c", &op);
  printf("Enter two operands: ");
  scanf("%lf %lf", &first, &second);

  switch (op) {
    case '+':
      printf("%.1lf + %.1lf = %.1lf", first, second, first + second);
      break;
    case '-':
      printf("%.1lf - %.1lf = %.1lf", first, second, first - second);
      break;
    case '*':
      printf("%.1lf * %.1lf = %.1lf", first, second, first * second);
      break;
    case '/':
      printf("%.1lf / %.1lf = %.1lf", first, second, first / second);
      break;
    // operator doesn't match any case constant
    default:
      printf("Error! operator is not correct");
  }
```

```c
  return 0;
}
```

Output :

```
Enter an operator (+, -, *,): *
Enter two operands: 1.5
4.5
1.5 * 4.5 = 6.8
```

ASSIGNMENT 11

Problem Statement :

Write a program in C to accept a number from the user and compute the following :

1) Square root of the number

2) Square of the number

3) Cube of the number

4) Check whether it's a prime number

5) Factorial of the number

6) Prime factors of the number

Code :

```c
#include <stdio.h>
#include <math.h> // Include the math library for sqrt()

int main() {
    double num, result;

    // Take user input for the number
    printf("Enter a number: ");
    scanf("%lf", &num);

    // Check if the number is negative
    if(num < 0) {
        printf("Square root of a negative number is imaginary\n");
        return 0;
    }

    // Calculate square root
    result = sqrt(num);

    // Display the result
    printf("Square Root of %.2lf is: %.2lf\n", num, result);

    return 0;
}
```

Output :

```
Enter a number: 16
Square Root of 16.00 is: 4.00
```

ASSIGNMENT 12

Problem Statement :

Write a program in C to accept 2 number from the user compute :

1) The smallest common divisor of the two numbers (other than 1, if any)

2) The GCD of the two numbers

Code :

```c
#include <stdio.h>

int main() {
    int a, b, i, gcd, scd = -1;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    // 1) Smallest Common Divisor (other than 1)
    for(i = 2; i <= a && i <= b; i++) {
        if(a % i == 0 && b % i == 0) {
            scd = i;
            break;
        }
    }

    if(scd == -1)
        printf("No common divisor other than 1\n");
    else
        printf("Smallest common divisor (other than 1): %d\n", scd);

    // 2) GCD (using loop method)
    for(i = 1; i <= a && i <= b; i++) {
        if(a % i == 0 && b % i == 0)
            gcd = i;
    }

    printf("GCD of the two numbers: %d\n", gcd);
```

```
    return 0;
}
```

Output :

```
Enter two numbers: 12 18
Smallest common divisor (other than 1): 2
GCD of the two numbers: 6
```

ASSIGNMENT 13

Problem Statement :

Write a program in C to accept a number from the user and print the digit of the number in reverse order.

Code :

```c
#include <stdio.h>

int main() {
  int n, reverse = 0, remainder, original;

  printf("Enter an integer: ");
  scanf("%d", &n);

  original = n;

  while (n != 0) {
    remainder = n % 10;
    reverse = reverse * 10 + remainder;
    n /= 10;
  }

  if (original % 10 == 0) {
    printf("Reversed number = %d", reverse);

    while (original % 10 == 0) {
      printf("0");
      original /= 10;
    }
  } else {
    printf("Reversed number = %d", reverse);
  }

  return 0;
}
```

Output :

```
Enter an integer: 2345
Reversed number = 5432
```

ASSIGNMENT 14

Problem Statement :

Write a program in C to input a binary number from the user and convert it into a decimal number.

Code :

```c
#include <stdio.h>

// function prototype
long long convert(long long);

int main() {

    long long n;

    printf("Enter a binary number: ");
    scanf("%lld", &n);

    printf("%lld in binary = %lld in decimal", n, convert(n));

    return 0;
}

// function definition
long long convert(long long n) {

    long long dec = 0;
    int i = 0, rem;

    while (n != 0) {

        // get remainder of n divided by 10
        rem = n % 10;
```

```
        // add the rem * (2 ^ i) to dec
        dec += rem << i; // Using bitwise shift instead of pow

        // divide n by 10
        n /= 10;

        // increment i
        ++i;
    }

    return dec;
}
```

Output :

```
Enter a binary number: 1101
1101 in binary = 13 in decimal
```

ASSIGNMENT 15

Problem Statement :

Write a program in C to generate pseudo random number.

Code :

```c
// C program to illustrate the use of rand() function
#include <stdio.h>
#include <stdlib.h>

int main()
{
    // Generate a random number using the rand() function
    int value = rand();

    // Print the generated random value
    printf("The Random Value is: %d", value);

    return 0;
}
```

Output :

```
The Random Value is: 1804289383
```

ASSIGNMENT 16

Problem Statement :

To accept a list of N integers and partition the list into two sublists containing even and odd numbers.

Code :

```c
#include <stdio.h>

int main() {
    int n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n], even[n], odd[n];
    int e = 0, o = 0;

    // Input elements
    printf("Enter %d integers:\n", n);
    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Partition into even and odd
    for(i = 0; i < n; i++) {
        if(arr[i] % 2 == 0) {
            even[e++] = arr[i];
        } else {
            odd[o++] = arr[i];
        }
    }
```

```
    // Display even numbers
    printf("Even numbers: ");
    for(i = 0; i < e; i++) {
        printf("%d ", even[i]);
    }

    // Display odd numbers
    printf("\nOdd numbers: ");
    for(i = 0; i < o; i++) {
        printf("%d ", odd[i]);
    }

    return 0;
}
```

Output :

```
Enter number of elements: 6
Enter 6 integers:
1 2 3 4 5 6
Even numbers: 2 4 6
Odd numbers: 1 3 5
```

ASSIGNMENT 17

Problem Statement :

To accept the number of terms and find the sum of the sine series.

Code :

```
#include <stdio.h>
#include <math.h>
#include <stdlib.h>

void main()
{
    int n, x1;
    float accuracy, term, denominator, x, sinx, sinval;

    printf("Enter the value of x (in degrees) \n");
    scanf("%f", &x);
    x1 = x;
    /*  Converting degrees to radians */
    x = x * (3.142 / 180.0);
    sinval = sin(x);
    printf("Enter the accuracy for the result \n");
    scanf("%f", &accuracy);
    term = x;
    sinx = term;
    n = 1;
    do
    {
        denominator = 2 * n * (2 * n + 1);
        term = -term * x * x / denominator;
        sinx = sinx + term;
        n = n + 1;
    } while (accuracy <= fabs(sinval - sinx));
    printf("Sum of the sine series = %f \n", sinx);
    printf("Using Library function sin(%d) = %f\n", x1, sin(x));
}
```

Output :

```
Enter the value of x (in degrees)
60
Enter the accuracy for the result
100
Sum of the sine series = 0.855862
Using Library function sin(60) = 0.866093
```

ASSIGNMENT 18

Problem Statement :

Write a C program that accepts a string from the user and performs the following string operations :

1) Calculate length of string

2) String reversal

3) Equality check of 2 strings

4) Check palindrome

5) Check substring

Code :

```c
#include <stdio.h>
#include <string.h>

int main() {
    char str1[100], str2[100], temp[100];
    int i, len, flag = 0;

    // Input first string
    printf("Enter first string: ");
    gets(str1);

    // 1) Calculate length
    len = strlen(str1);
    printf("Length of string: %d\n", len);

    // 2) String reversal
    for(i = 0; i < len; i++) {
        temp[i] = str1[len - i - 1];
    }
    temp[len] = '\0';
    printf("Reversed string: %s\n", temp);

    // Input second string for comparison
    printf("Enter second string: ");
    gets(str2);
```

```
    // 3) Equality check
    if(strcmp(str1, str2) == 0)
        printf("Strings are equal\n");
    else
        printf("Strings are not equal\n");

    // 4) Check palindrome
    if(strcmp(str1, temp) == 0)
        printf("String is a palindrome\n");
    else
        printf("String is not a palindrome\n");

    // 5) Check substring
    if(strstr(str1, str2) != NULL)
        printf("Second string is a substring of first string\n");
    else
        printf("Second string is not a substring of first string\n");

    return 0;
}
```

Output :

```
Enter first string: madam
Length of string: 5
Reversed string: madam

Enter second string: mad
Strings are not equal
String is a palindrome
Second string is a substring of first string
```

ASSIGNMENT 19

Problem Statement :

Create structure employee for string details (name, designation, gender, date of joining and salary). Define function members to compute :

1) Total number of employees in an organisation

2) Count of male and female employees

3) Employees with salary more than 10000

4) Employees with designation "Asst Manager"

Code :

```c
#include <stdio.h>
#include <string.h>

struct employee {
    char name[50];
    char designation[50];
    char gender[10];
    char doj[15]; // date of joining
    float salary;
};

// Function to count total employees
int totalEmployees(int n) {
    return n;
}

// Function to count male and female employees
void countGender(struct employee emp[], int n) {
    int male = 0, female = 0;
    for(int i = 0; i < n; i++) {
        if(strcmp(emp[i].gender, "male") == 0)
            male++;
        else if(strcmp(emp[i].gender, "female") == 0)
            female++;
    }
    printf("Male Employees: %d\n", male);
    printf("Female Employees: %d\n", female);
}
```

```c
// Employees with salary > 10000
void highSalary(struct employee emp[], int n) {
    printf("Employees with salary > 10000:\n");
    for(int i = 0; i < n; i++) {
        if(emp[i].salary > 10000)
            printf("%s\n", emp[i].name);
    }
}

// Employees with designation "Asst Manager"
void asstManager(struct employee emp[], int n) {
    printf("Employees with designation Asst Manager:\n");
    for(int i = 0; i < n; i++) {
        if(strcmp(emp[i].designation, "Asst Manager") == 0)
            printf("%s\n", emp[i].name);
    }
}

int main() {
    int n;
    printf("Enter number of employees: ");
    scanf("%d", &n);

    struct employee emp[n];
```

```c
    for(int i = 0; i < n; i++) {
        printf("\nEnter details of employee %d:\n", i+1);
        printf("Name: ");
        scanf("%s", emp[i].name);
        printf("Designation: ");
        scanf("%s", emp[i].designation);
        printf("Gender (male/female): ");
        scanf("%s", emp[i].gender);
        printf("Date of Joining: ");
        scanf("%s", emp[i].doj);
        printf("Salary: ");
        scanf("%f", &emp[i].salary);
    }

    printf("\nTotal Employees: %d\n", totalEmployees(n));

    countGender(emp, n);
    highSalary(emp, n);
    asstManager(emp, n);

    return 0;
}
```

Output :

```
Enter number of employees: 3

Enter details of employee 1:
Name: Ravi
Designation: Manager
Gender: male
Date of Joining: 12-01-2020
Salary: 15000

Enter details of employee 2:
Name: Priya
Designation: Asst Manager
Gender: female
Date of Joining: 05-03-2021
Salary: 12000

Enter details of employee 3:
Name: Aman
Designation: Clerk
Gender: male
Date of Joining: 10-07-2022
Salary: 8000

Total Employees: 3
Male Employees: 2
Female Employees: 1
Employees with salary > 10000:
Ravi
Priya
Employees with designation Asst Manager:
Priya
```

ASSIGNMENT 20

Problem Statement :

Write a C program to swap two numbers with and without using pointers.

Code :

```c
#include <stdio.h>

void swap(int a, int b) {
    int temp;
    temp = a;
    a = b;
    b = temp;
    printf("Inside function (after swap): a = %d, b = %d\n", a, b);
}

int main() {
    int x, y;
    printf("Enter two numbers: ");
    scanf("%d %d", &x, &y);

    swap(x, y);

    printf("In main (after function call): x = %d, y = %d\n", x, y);
    return 0;
}
```

```c
#include <stdio.h>

void swap(int *a, int *b) {
    int temp;
    temp = *a;
    *a = *b;
    *b = temp;
}

int main() {
    int x, y;
    printf("Enter two numbers: ");
    scanf("%d %d", &x, &y);

    swap(&x, &y);

    printf("After swapping: x = %d, y = %d\n", x, y);
    return 0;
}
```

Output :

```
Enter two numbers: 5 10
Inside function (after swap): a = 10, b = 5
In main (after function call): x = 5, y = 10
```

```
Enter two numbers: 5 10
After swapping: x = 10, y = 5
```

ASSIGNMENT 21

Problem Statement :

Write a c program to copy contents of one file to another using File handling.

Code :

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    FILE *source, *destination;
    char sourceFile[100], destFile[100];
    char ch;

    // Input source file name
    printf("Enter source file name: ");
    scanf("%s", sourceFile);

    // Open source file in read mode
    source = fopen(sourceFile, "r");
    if (source == NULL) {
        printf("Error: Cannot open source file.\n");
        exit(1);
    }

    // Input destination file name
    printf("Enter destination file name: ");
    scanf("%s", destFile);

    // Open destination file in write mode
    destination = fopen(destFile, "w");
    if (destination == NULL) {
        printf("Error: Cannot open destination file.\n");
        fclose(source);
        exit(1);
    }
```

```
    // Copy contents
    while ((ch = fgetc(source)) != EOF) {
        fputc(ch, destination);
    }

    printf("File copied successfully!\n");

    // Close files
    fclose(source);
    fclose(destination);

    return 0;
}
```

Output :

```
Enter source file name: input.txt
Enter destination file name: output.txt
File copied successfully!
```

ASSIGNMENT 22

Problem Statement :

Write a menu driven program in C to perform various string operations using in-built function.

Code :

```c
#include <stdio.h>
#include <string.h>

int main() {
    char str1[100], str2[100], temp[100];
    int choice, len;

    printf("Enter first string: ");
    scanf("%s", str1);

    printf("Enter second string: ");
    scanf("%s", str2);

    do {
        printf("\n----- MENU -----\n");
        printf("1. Find Length of String\n");
        printf("2. Copy String\n");
        printf("3. Concatenate Strings\n");
        printf("4. Compare Strings\n");
        printf("5. Reverse String\n");
        printf("6. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
```

```c
        switch(choice) {
            case 1:
                len = strlen(str1);
                printf("Length of string: %d\n", len);
                break;

            case 2:
                strcpy(temp, str1);
                printf("Copied string: %s\n", temp);
                break;

            case 3:
                strcpy(temp, str1);
                strcat(temp, str2);
                printf("Concatenated string: %s\n", temp);
                break;

            case 4:
                if(strcmp(str1, str2) == 0)
                    printf("Strings are equal\n");
                else
                    printf("Strings are not equal\n");
                break;

            case 5:
                strcpy(temp, str1);
                strrev(temp);  // Note: works in some compilers like Turbo C
                printf("Reversed string: %s\n", temp);
                break;

            case 6:
                printf("Exiting program...\n");
                break;
```

```c
            default:
                printf("Invalid choice!\n");
        }

    } while(choice != 6);

    return 0;
}
```

Output :

```
Enter first string: hello
Enter second string: world

----- MENU -----
1. Find Length of String
2. Copy String
3. Concatenate Strings
4. Compare Strings
5. Reverse String
6. Exit
Enter your choice: 1
Length of string: 5

Enter your choice: 3
Concatenated string: helloworld

Enter your choice: 4
Strings are not equal

Enter your choice: 6
Exiting program...
```